But

$$\text{Var}(X_i) = E[X_{2i}] - (E[X_i])^2 = E[X_i] - (E[X_i])^2 \quad \text{since } X_{2i} = X_i = p - p^2$$

Thus,

$$\text{Var}(X) = np(1-p)$$

Example 4c Sampling from a finite population

Consider a set of N people, each of whom has an opinion about a certain subject that is measured by a real number v that represents the person's "strength of feeling" about the subject. Let vi represent the strength of feeling of person i,i=1,…N.

Suppose that the quantities vi,i=1,…,N, are unknown and, to gather information, a group of n of the N people is "randomly chosen" in the sense that all of the (N2) subsets of size n are equally likely to be chosen. These n people are then questioned and their feelings determined. If S denotes the sum of the n sampled values, determine its mean and variance.

An important application of the preceding problem is to a forthcoming election in which each person in the population is either for or against a certain candidate or proposition. If we take vi to equal 1 if person i is in favor and 0 if he or she is against, then $\bar{v} = \sum_{i=1}^{N} v_i / N$ represents the proportion of the population that is in favor. To estimate $\bar{v}$, a random sample of n people is chosen, and these people are polled. The proportion of those polled who are in favor – that is, *S/n* – is often used as an estimate of $\bar{v}$.

Solution

For each person i,i=1,…,N, define an indicator variable Ii to indicate whether or not that person is included in the sample. That is,

$$I_i = \begin{cases} 1 & \text{if person } i \text{ is in the random sample} \\ 0 & \text{otherwise} \end{cases}$$

Now,S can be expressed by

$$S = \sum_{i=1}^{N} v_i I_i$$

so

$$E[S] = \sum_{i=1}^{N} v_i E[I_i] \quad \text{Var}(S) = \sum_{i=1}^{N} \text{Var}(v_i I_i) + 2\sum\sum_{i<j} \text{Cov}(v_i I_i, v_j I_j) = \sum_{i=1}^{N} v_i^2 \text{Var}(I_i) + 2\sum\sum_{i<j} v_i v_j \text{Cov}(I_i, I_j)$$

Because

$$E[I_i] = \frac{n}{N} \quad E[I_i I_j] = \frac{n}{N}\frac{n-1}{N-1}$$

it follows that

$$\text{Var}(I_i) = \frac{n}{N}\left(1 - \frac{n}{N}\right) \quad \text{Cov}(I_i, I_j) = \frac{n(n-1)}{N(N-1)} - \left(\frac{n}{N}\right)^2 = -\frac{n(N-n)}{N^2(N-1)}$$

Hence,

$$E[S] = n \sum_{i=1}^{N} v_i / N = n\bar{v} \quad \text{Var}(S) = \frac{n}{N}\left(\frac{N-n}{N}\right) \sum_{i=1}^{N} v_i^2 - 2\frac{n}{N}\left(\frac{N-n}{N}\right)\frac{1}{2(N-1)} \sum\sum_{i<j} v_i v_j$$

The expression for Var (S) can be simplified somewhat by using the identity $(v_1 + \cdots + v_N)^2 = \sum_{i=1}^{N} v_i^2 + 2\sum\sum_{i<j} v_i v_j$. After some simplification, we obtain

$$\text{Var}(S) = \frac{n(N-n)}{N-1}\left(\sum_{i=1}^{N} v_i^2 / N - \bar{v}^2\right)$$

Consider now the special case in which *Np* of the v's are equal to 1 and the remainder equal to 0. Then, in this case,S is a hypergeometric random variable and has mean and variance given, respectively, by

$$E[S] = n\bar{v} = np \quad \text{since } \bar{v} = Np/N = p$$

and

$$\text{Var}(S) = \frac{n(N-n)}{N-1}\left(\frac{Np}{N} - p^2\right) = \frac{n(N-n)}{N-1}p(1-p)$$

The quantity *S/n*, equal to the proportion of those sampled that have values equal to 1, is such that

$$E[S_n]=p \quad \text{Var}(S_n)=\frac{N-n}{n(N-1)}p(1-p)$$

The correlation of two random variables X and Y, denoted by ρ(X,Y), is defined, as long as Var (X)Var (Y) is positive, by

$$\rho(X,Y)=\frac{\text{Cov}(X,Y)}{\sqrt{\text{Var}(X)\,\text{Var}(Y)}}$$

It can be shown that

$$-1\le\rho(X,Y)\le 1 \tag{4.2}$$

To prove **Equation (4.2)**, suppose that and Y have variances given by σ2x and σ2y, respectively. Then, on the one hand,

$$0\le\text{Var}(X\sigma x+Y\sigma y)=\text{Var}(X)\sigma^2x+\text{Var}(Y)\sigma^2y+2\,\text{Cov}(X,Y)\sigma x\sigma y=2[1+\rho(X,Y)]$$

implying that

$$-1\le\rho(X,Y)$$

On the other hand,

$$0\le\text{Var}(X\sigma x-Y\sigma y)=\text{Var}(X)\sigma^2x+\text{Var}(Y)(-\sigma y)^2-2\,\text{Cov}(X,Y)\sigma x\sigma y=2[1+\rho(X,Y)]$$

implying that

$$\rho(X,Y)\le 1$$

which completes the proof of **Equation (4.2)**.

In fact, since Var (Z)=0 implies that Z is constant with probability 1 (this intuitive relationship will be rigorously proven in **Chapter 8**), it follows from the proof of **Equation (4.2)** that ρ(X,Y)=1 implies that Y=a+bX, where b=σy/σx>0 and ρ(X,Y)=−1 implies that Y=a+bX, where b=−σy/σx<0. We leave it as an exercise for the reader to show that the reverse is also true: that if Y=a+bX, then ρ(X,Y) is either +1 or −1, depending on the sign of b.

The correlation coefficient is a measure of the degree of linearity between X and Y. A value of ρ(X,Y) near +1 or −1 indicates a high degree of linearity between X and Y, whereas a value near 0 indicates that such linearity is absent. A positive value of ρ(X,Y) indicates that Y tends to increase when X does, whereas a negative value indicates that Y tends to decrease when X increases. If ρ(X,Y)=0, then X and Y are said to be *uncorrelated*.

Example 4d

Let IA and IB be indicator variables for the events A and B. That is,

$$I_A=\begin{cases}1 & \text{if } A \text{ occurs}\\ 0 & \text{otherwise}\end{cases} \qquad I_B=\begin{cases}1 & \text{if } B \text{ occurs}\\ 0 & \text{otherwise}\end{cases}$$

Then

$$E[I_A]=P(A)\quad E[I_B]=P(B)\quad E[I_AI_B]=P(AB)$$

so

$$\text{Cov}(I_A,I_B)=P(AB)-P(A)P(B)=P(B)[P(A|B)-P(A)]$$

Thus, we obtain the quite intuitive result that the indicator variables for A and B are either positively correlated, uncorrelated, or negatively correlated, depending on whether P(A|B) is, respectively, greater than, equal to, or less than P(A).

Our next example shows that the sample mean and a deviation from the sample mean are uncorrelated.

Example 4e

Let X1,…,Xn be independent and identically distributed random variables having variance σ2. Show that

$$\text{Cov}(X_i-\bar{X},\bar{X})=0$$

Solution

We have

$$\text{Cov}(X_i - \bar{X}, \bar{X}) = \text{Cov}(X_i, \bar{X}) - \text{Cov}(\bar{X}, \bar{X}) = \text{Cov}\left(X_i, \frac{1}{n}\sum_{j=1}^{n} X_j\right) - \text{Var}(\bar{X}) = \frac{1}{n}\sum_{j=1}^{n}\text{Cov}(X_i, X_j) - \frac{\sigma^2}{n} = \frac{\sigma^2}{n} - \frac{\sigma^2}{n} = 0$$

where the next-to-last equality uses the result of **Example 4a** and the final equality follows because

$$\text{Cov}(X_i, X_j) = \begin{cases} 0 & \text{if } j \neq i \text{ by independence} \\ \sigma^2 & \text{if } j = i \text{ since } \text{Var}(X_i) = \sigma^2 \end{cases}$$

Although $\bar{X}$ and the deviation $X_i - \bar{X}$ are uncorrelated, they are not, in general, independent. However, in the special case where the X_i are normal random variables, it turns out that not only is $\bar{X}$ independent of a single deviation, but it is independent of the entire sequence of deviations $X_j - \bar{X}, j = 1, \ldots, n$. This result will be established in **Section 7.8**, where we will also show that, in this case, the sample mean $\bar{X}$ and the sample variance S^2 are independent, with $(n-1)S^2/\sigma^2$ having a chi-squared distribution with $n-1$ degrees of freedom. (See **Example 4a** for the definition of S^2.)

Example 4f

Consider m independent trials, each of which results in any of r possible outcomes with probabilities $p_1, \ldots, p_r, \sum_{i=1}^{r} p_i = 1$. If we let $N_i, i = 1, \ldots, r$, denote the number of the m trials that result in outcome i, then $N_1, N_2, \ldots, N_r$ have the multinomial distribution

$$P\{N_1 = n_1, \ldots, N_r = n_r\} = \frac{m!}{n_1! \ldots n_r!} p_1^{n_1} \cdots p_r^{n_r}, \quad \sum_{i=1}^{r} n_i = m$$

For $i \neq j$, it seems likely that when N_i is large, N_j would tend to be small; hence, it is intuitive that they should be negatively correlated. Let us compute their covariance by using **Proposition 4.2**(iv) and the representation

$$N_i = \sum_{k=1}^{m} I_i(k) \qquad \text{and} \qquad N_j = \sum_{k=1}^{m} I_j(k)$$

where

$$I_i(k) = \begin{cases} 1 & \text{if trial } k \text{ results in outcome } i \\ 0 & \text{otherwise} \end{cases} \qquad I_j(k) = \begin{cases} 1 & \text{if trial } k \text{ results in outcome } j \\ 0 & \text{otherwise} \end{cases}$$

From **Proposition 4.2**(iv), we have

$$\text{Cov}(N_i, N_j) = \sum_{\ell=1}^{m}\sum_{k=1}^{m} \text{Cov}(I_i(k), I_j(\ell))$$

Now, on the one hand, when $k \neq \ell$,

$$\text{Cov}(I_i(k), I_j(\ell)) = 0$$

since the outcome of trial k is independent of the outcome of trial ℓ. On the other hand,

$$\text{Cov}(I_i(\ell), I_j(\ell)) = E[I_i(\ell) I_j(\ell)] - E[I_i(\ell)]E[I_j(\ell)] = 0 - p_i p_j = -p_i p_j$$

where the equation uses the fact that $I_i(\ell) I_j(\ell) = 0$, since trial ℓ cannot result in both outcome i and outcome j. Hence, we obtain

$$\text{Cov}(N_i, N_j) = -m p_i p_j$$

which is in accord with our intuition that N_i and N_j are negatively correlated.

7.5 Conditional Expectation

7.5.1 Definitions

Recall that if X and Y are jointly discrete random variables, then the conditional probability mass function of X, given that Y=y, is defined for all y such that P{Y=y}>0, by

pX|Y(x|y)=P{X=x|Y=y}=p(x,y)pY(y)

It is therefore natural to define, in this case, the conditional expectation of X given that Y=y, for all values of y such that pY(y)>0, by

E[X|Y=y]=∑x xP{X=x|Y=y}=∑x xpX|Y(x|y)

> ***Example 5a***
>
> If X and Y are independent binomial random variables with identical parameters n and p, calculate the conditional expected value of X given that X+Y=m.
>
> Solution
>
> Let us first calculate the conditional probability mass function of X given that X+Y=m. For k≤min(n,m),
>
> P{X=k|X+Y=m}=P{ X=k,X+Y=m }P{ X+Y=m }=P{ X=k,Y=m−k }P{ X+Y=m }=P{ X=k }P{ Y=m−k }P{ X+Y=m }=(nk) pk(1−p)n−k (nm−k) pm−k(1−p)n−m+k(2nm) pm(1−p)2n−m=(nk) (nm−k)(2nm)
>
> where we have used the fact (see **Example 3f** of **Chapter 6**) that X+Y is a binomial random variable with parameters 2n and p. Hence, the conditional distribution of X, given that X+Y=m, is the hypergeometric distribution, and from **Example 2g**, we obtain
>
> E[X|X+Y=m]=m2

Similarly, let us recall that if X and Y are jointly continuous with a joint probability density function f(x,y), then the conditional probability density of X, given that Y=y, is defined for all values of y such that fY(y)>0 by

fX|Y(x|y)=f(x,y)fY(y)

It is natural, in this case, to define the conditional expectation of X, given that Y=y, by

E[X|Y=y]=∫−∞∞xfX|Y(x|y) dx

provided that fY(y)>0.

> ***Example 5b***
>
> Suppose that the joint density of X and Y is given by
>
> f(x,y)=e−x/ye−yy 0<x<∞, 0<y<∞
>
> Compute E[X|Y=y].
>
> Solution
>
> We start by computing the conditional density
>
> fX|Y(x|y)=f(x,y)fY(y)=f(x,y)∫−∞∞f(x,y) dx=(1/y)e−x/ye−y∫0∞(1/y)e−x/ye−y dx=(1/y)e−x/y∫0∞(1/y)e−x/y dx=1ye−x/y
>
> Hence, the conditional distribution of X, given that Y=y, is just the exponential distribution with mean y. Thus,
>
> E[X|Y=y]=∫0∞xye−x/y dx=y

Remark Just as conditional probabilities satisfy all of the properties of ordinary probabilities, so do conditional expectations satisfy the properties of ordinary expectations. For instance, such formulas as

$$E[g(X)|Y=y]=\begin{cases}\sum_x g(x)\, p_{X|Y}(x|y) & \text{in the discrete case}\\ \int_{-\infty}^{\infty} g(x)\, f_{X|Y}(x|y)\, dx & \text{in the continuous case}\end{cases}$$

and

$$E\left[\sum_{i=1}^{n} X_i|Y=y\right]=\sum_{i=1}^{n} E[X_i|Y=y]$$

remain valid. As a matter of fact, conditional expectation given that Y=y can be thought of as being an ordinary expectation on a reduced sample space consisting only of outcomes for which Y=y.

7.5.2 Computing expectations by conditioning

Let us denote by E[X|Y] that function of the random variable Y whose value at Y=y is E[X|Y=y]. Note that E[X|Y] is itself a random variable. An extremely important property of conditional expectations is given by the following proposition.

Proposition 5.1 The conditional expectation formula

$$E[X]=E[E[X|Y]] \tag{5.1}$$

If Y is a discrete random variable, then **Equation (5.1)** states that

$$E[X]=\sum_y E[X|Y=y]P\{Y=y\} \tag{5.1a}$$

whereas if Y is continuous with density $f_Y(y)$, then **Equation (5.1)** states

$$E[X]=\int_{-\infty}^{\infty} E[X|Y=y]\, f_Y(y)\, dy \tag{5.1b}$$

We now give a proof of **Equation (5.1)** in the case where X and Y are both discrete random variables.

Proof of Equation (5.1) When X and Y Are Discrete: We must show that

$$E[X]=\sum_y E[X|Y=y]P\{Y=y\} \tag{5.2}$$

Now, the right-hand side of **Equation (5.2)** can be written as

$$\sum_y E[X|Y=y]P\{Y=y\}=\sum_y\sum_x x P\{X=x|Y=y\}P\{Y=y\}=\sum_y\sum_x x P\{X=x,Y=y\}P\{Y=y\}P\{Y=y\}=\sum_y\sum_x x P\{X=x,Y=y\}=\sum_x x\sum_y P\{X=x,Y=y\}=\sum_x x P\{X=x\}=E[X]$$

and the result is proved.

One way to understand **Equation (5.2)** is to interpret it as follows: To calculate E[X], we may take a weighted average of the conditional expected value of X given that Y=y, each of the terms E[X|Y=y] being weighted by the probability of the event on which it is conditioned. (Of what does this remind you?) This is an extremely useful result that often enables us to compute expectations easily by first conditioning on some appropriate random variable. The following examples illustrate its use.

Example 5c

A miner is trapped in a mine containing 3 doors. The first door leads to a tunnel that will take him to safety after 3 hours of travel. The second door leads to a tunnel that will return him to the mine after 5 hours of travel. The third door leads to a tunnel that will return him to the mine after 7 hours. If we assume that the miner is at all times equally likely to choose any one of the doors, what is the expected length of time until he reaches safety?

Solution

Let X denote the amount of time (in hours) until the miner reaches safety, and let Y denote the door he initially chooses. Now,

$$E[X] = E[X|Y=1]P\{Y=1\}+E[X|Y=2]P\{Y=2\}+E[X|Y=3]P\{Y=3\} = \frac{1}{3}(E[X|Y=1]+E[X|Y=2]+E[X|Y=3])$$

However,

$$E[X|Y=1]=3E[X|Y=2]=5+E[X]E[X|Y=3]=7+E[X] \quad (5.3)$$

To understand why **Equation (5.3)** is correct, consider, for instance,E[X|Y=2] and reason as follows: If the miner chooses the second door, he spends 5 hours in the tunnel and then returns to his cell. But once he returns to his cell, the problem is as before; thus, his expected additional time until safety is just E[X]. Hence,E[X|Y=2]=5+E[X]. The argument behind the other equalities in **Equation (5.3)** is similar. Hence,

$$E[X]=13(3+5+E[X]+7+E[X])$$

or

$$E[X]=15$$

Example 5d Expectation of a sum of a random number of random variables

Suppose that the number of people entering a department store on a given day is a random variable with mean 50. Suppose further that the amounts of money spent by these customers are independent random variables having a common mean of $8. Finally, suppose also that the amount of money spent by a customer is also independent of the total number of customers who enter the store. What is the expected amount of money spent in the store on a given day?

Solution

If we let N denote the number of customers who enter the store and Xi the amount spent by the ith such customer, then the total amount of money spent can be expressed as $\sum_{i=1}^{N} X_i$. Now,

$$E\left[\sum_{1}^{N} X_i\right]=E\left[E\left[\sum_{1}^{N} X_i|N\right]\right]$$

But

$$E\left[\sum_{1}^{N} X_i|N=n\right] = E\left[\sum_{1}^{n} X_i|N=n\right] = E\left[\sum_{1}^{n} X_i\right] \text{ by the independence of the } X_i \text{ and } N = nE[X] \text{ where } E[X]=E[X_i]$$

which implies that

$$E\left[\sum_{1}^{N} X_i|N\right]=NE[X]$$

Thus,

$$E\left[\sum_{i=1}^{N} X_i\right]=E[NE[X]]=E[N]E[X]$$

Hence, in our example, the expected amount of money spent in the store is 50×$8, or $400.

Example 5e

The game of craps is begun by rolling an ordinary pair of dice. If the sum of the dice is 2, 3, or 12, the player loses. If it is 7 or 11, the player wins. If it is any other number i, the player continues to roll the dice until the sum is either 7 or i. If it is 7, the player loses; if it is i, the player wins. Let R denote the number of rolls of the dice in a game of craps. Find -5pt

a. E[R];
b. E[R| player wins];
c. E[R| player loses].

Solution

If we let Pi denote the probability that the sum of the dice is i, then

$$P_i=P_{14-i}=i-136, \quad i=2,\ldots,7$$

To compute E[R], we condition on S, the initial sum, giving

$$E[R]=\sum_{i=2}^{12}E[R|S=i]P_i$$

However,

$$E[R|S=i]=\begin{cases}1, & \text{if } i=2,3,7,11,12\\ 1+P_i+P_7, & \text{otherwise}\end{cases}$$

The preceding equation follows because if the sum is a value i that does not end the game, then the dice will continue to be rolled until the sum is either i or 7, and the number of rolls until this occurs is a geometric random variable with parameter P_i+P_7. Therefore,

$$E[R]=1+\sum_{i=4}^{6}\frac{P_i}{P_i+P_7}+\sum_{i=8}^{10}\frac{P_i}{P_i+P_7}=1+2(3/9+4/10+5/11)=3.376$$

To determine E[R| win], let us start by determining p, the probability that the player wins. Conditioning on S yields

$$p=\sum_{i=2}^{12}P\{\text{win}|S=i\}P_i=P_7+P_{11}+\sum_{i=4}^{6}\frac{P_i}{P_i+P_7}P_i+P_7P_i+\sum_{i=8}^{10}\frac{P_i}{P_i+P_7}P_i+P_7P_i=0.493$$

where the preceding uses the fact that the probability of obtaining a sum of i before one of 7 is $P_i/(P_i+P_7)$. Now, let us determine the conditional probability mass function of S, given that the player wins. Letting $Q_i=P\{S=i|\text{ win}\}$, we have

$$Q_2=Q_3=Q_{12}=0,\quad Q_7=P_7/p,\quad Q_{11}=P_{11}/p$$

and, for i=4,5,6,8,9,10,

$$Q_i=P\{S=i,\text{ win}\}P\{\text{win}\}=P_iP\{\text{win}|S=i\}p=\frac{P_i}{2}\frac{p}{(P_i+P_7)}$$

Now, conditioning on the initial sum gives

$$E[R|\text{ win }]=\sum_i E[R|\text{ win },S=i]Q_i$$

However, as was noted in **Example 2j** of **Chapter 6**, given that the initial sum is i, the number of additional rolls needed and the outcome (whether a win or a loss) are independent. (This is easily seen by first noting that conditional on an initial sum of i, the outcome is independent of the number of additional dice rolls needed and then using the symmetry property of independence, which states that if event A is independent of event B, then event B is independent of event A.) Therefore,

$$E[R|\text{win }]=\sum_i E[R|S=i]Q_i=1+\sum_{i=4}^{6}\frac{Q_iP_i}{P_i+P_7}+\sum_{i=8}^{10}\frac{Q_iP_i}{P_i+P_7}=2.938$$

Although we could determine E[R| player loses] exactly as we did E[R| player wins], it is easier to use

$$E[R]=E[R|\text{ win }]p+E[R|\text{ lose }](1-p)$$

implying that

$$E[R|\text{ lose }]=\frac{E[R]-E[R|\text{ win }]p}{1-p}=3.801$$

Example 5f

As defined in **Example 5d** of **Chapter 6**, the bivariate normal joint density function of the random variables X and Y is

$$f(x,y)=\frac{1}{2\pi\sigma_x\sigma_y\sqrt{1-\rho^2}}\exp\left\{-\frac{1}{2(1-\rho^2)}\left[\left(\frac{x-\mu_x}{\sigma_x}\right)^2+\left(\frac{y-\mu_y}{\sigma_y}\right)^2-2\rho\frac{(x-\mu_x)(y-\mu_y)}{\sigma_x\sigma_y}\right]\right\}$$

We will now show that ρ is the correlation between X and Y. As shown in **Example 5c**, $\mu_x=E[X],\sigma_x^2=\text{Var}(X)$, and $\mu_y=E[Y],\sigma_y^2=\text{Var}(Y)$. Consequently,

$$\text{Corr}(X,Y)=\frac{\text{Cov}(X,Y)}{\sigma_x\sigma_y}=\frac{E[XY]-\mu_x\mu_y}{\sigma_x\sigma_y}$$

To determine E[XY], we condition on Y. That is, we use the identity

$$E[XY]=E[E[XY|Y]]$$

Recalling from **Example 5d** that the conditional distribution of X given that Y=y is normal with mean μx+ρσxσy(y−μy), we see that

$$E[XY|Y=y]=E[Xy|Y=y]=yE[X|Y=y]=y[\mu x+\rho\sigma x\sigma y(y-\mu y)]=y\mu x+\rho\sigma x\sigma y(y2-\mu yy)$$

Consequently,

$$E[XY|Y]=Y\mu x+\rho\sigma x\sigma y(Y2-\mu yY)$$

implying that

$$E[XY]=E[Y\mu x+\rho\sigma x\sigma y(Y2-\mu yY)]=\mu xE[Y]+\rho\sigma xE[Y2-\mu yY]=\mu x\mu y+\rho\sigma x\sigma y(E[Y2]-\mu 2y)=\mu x\mu y+\rho\sigma x\sigma y \text{ Var } (Y)=\mu x\mu y+\rho\sigma x\sigma y$$

Therefore,

$$\text{Corr } (X,Y)=\rho\sigma x\sigma y\sigma x\sigma y=\rho$$

Sometimes E[X] is easy to compute, and we use the conditioning identity to compute a conditional expected value. This approach is illustrated by our next example.

Example 5g

Consider n independent trials, each of which results in one of the outcomes 1,…,k, with respective probabilities p1,…,pk,∑i=1kpi=1. Let Ni denote the number of trials that result in outcome i,i=1,…,k. For i≠j, find

(a) $E[Nj|Ni>0]$ and (b) $E[Nj|Ni>1]$

Solution

To solve (a), let

$$I=\begin{cases}0, & \text{if } Ni=01,\\ 1, & \text{if } Ni>0\end{cases}$$

Then

$$E[Nj]=E[Nj|I=0]P\{I=0\}+E[Nj|I=1]P\{I=1\}$$

or, equivalently,

$$E[Nj]=E[Nj|Ni=0]P\{Ni=0\}+E[Nj|Ni>0]P\{Ni>0\}$$

Now, the unconditional distribution of Nj is binomial with parameters n,pj. Also, given that Ni=r, each of the n−r trials that does not result in outcome i will, independently, result in outcome j with probability P(j| not i)=pj1−pi. Consequently, the conditional distribution of Nj, given that Ni=r, is binomial with parameters n−r,pj1−pi. (For a more detailed argument for this conclusion, see **Example 4c** of **Chapter 6**.) Because P{Ni=0}=(1−pi)n, the preceding equation yields

$$npj=npj1-pi(1-pi)n+E[Nj|Ni>0](1-(1-pi)n)$$

giving the result

$$E[Nj|Ni>0]=npj\ 1-(1-pi)n-11-(1-pi)n$$

We can solve part (b) in a similar manner. Let

$$J=\begin{cases}0, & \text{if } Ni=01,\\ 1, & \text{if } Ni=12,\\ 2, & \text{if } Ni>1\end{cases}$$

Then

$$E[Nj]=E[Nj|J=0]P\{J=0\}+E[Nj|J=1]P\{J=1\}\ +E[Nj|J=2]P\{J=2\}$$

or, equivalently,

$$E[N_j]=E[N_j|N_i=0]P\{N_i=0\}+E[N_j|N_i=1]P\{N_i=1\} \quad +E[N_j|N_i>1]P\{N_i>1\}$$

This equation yields

$$np_j=np_j1-p_i(1-p_i)n+(n-1)p_j1-p_inp_i(1-p_i)n-1 \quad +E[N_j|N_i>1](1-(1-p_i)n-np_i(1-p_i)n-1)$$

giving the result

$$E[N_j|N_i>1]=np_j[1-(1-p_i)n-1-(n-1)p_i(1-p_i)n-2]1-(1-p_i)n-np_i(1-p_i)n-1$$

It is also possible to obtain the variance of a random variable by conditioning. We illustrate this approach by the following example.

Example 5h Variance of the geometric distribution

Independent trials, each resulting in a success with probability p, are successively performed. Let N be the time of the first success. Find Var (N).

Solution

Let Y=1 if the first trial results in a success and Y=0 otherwise. Now,

$$\text{Var}(N)=E[N^2]-(E[N])2$$

To calculate E[N2], we condition on Y as follows:

$$E[N^2]=E[E[N^2|Y]]$$

However,

$$E[N^2|Y=1]=1E[N^2|Y=0]=E[(1+N)2]$$

These two equations follow because, on the one hand, if the first trial results in a success, then, clearly,N=1; thus,N2=1. On the other hand, if the first trial results in a failure, then the total number of trials necessary for the first success will have the same distribution as 1 (the first trial that results in failure) plus the necessary number of additional trials. Since the latter quantity has the same distribution as N, we obtain $E[N^2|Y=0]=E[(1+N)2]$Hence,

$$E[N^2]=E[N^2|Y=1]P\{Y=1\}+E[N^2|Y=0]P\{Y=0\}=p+(1-p)E[(1+N)2]=1+(1-p)E[2N+N^2]$$

However, as was shown in **Example 8b** of **Chapter 4**,E[N]=1/p; therefore,

$$E[N^2]=1+2(1-p)p+(1-p)E[N^2]$$

or

$$E[N^2]=2-pp2$$

Consequently,

$$\text{Var}(N)=E[N^2]-(E[N])2=2-pp2-(1p)2=1-pp2$$

Example 5i

Consider a gambling situation in which there are r players, with player i initially having ni units, ni>0, i=1, … , r. At each stage, two of the players are chosen to play a game, with the winner of the game receiving 1 unit from the loser. Any player whose fortune drops to 0 is eliminated, and this continues until a single player has all $n\equiv\sum i=1rni$ units, with that player designated as the victor. Assuming that the results of successive games are independent and that each game is equally likely to be won by either of its two players, find the average number of stages until one of the players has all n units.

Solution

To find the expected number of stages played, suppose first that there are only 2 players, with players 1 and 2 initially having j and n−j units, respectively. Let Xj denote the number of stages that will be played, and let mj=E[Xj].Then, for j=1,...,n−1,

Xj=1+Aj

where Aj is the additional number of stages needed beyond the first stage. Taking expectations gives

mj=1+E[Aj]

Conditioning on the result of the first stage then yields

mj=1+E[Aj|1 wins first stage]1/2+E[Aj|2 wins first stage]1/2

Now, if player 1 wins at the first stage, then the situation from that point on is exactly the same as in a problem that supposes that player *1* starts with j+1 and player *2* with n−(j+1) units. Consequently,

E[Aj|1 wins first stage]=mj+1

and, analogously,

E[Aj|2 wins first stage]=mj−1

Thus,

mj=1+12 mj+1+12 mj−1

or, equivalently,

$$m_{j+1}=2m_j-m_{j-1}-2, \quad j=1,\ldots,n-1 \tag{5.4}$$

Using that m0=0, the preceding equation yields

m2=2m1−2m3=2m2−m1−2=3m1−6=3(m1−2)m4=2m3−m2−2=4m1−12=4(m1−3)

suggesting that

$$m_i=i(m_1-i+1), \qquad i=1,\ldots,n \tag{5.5}$$

To prove the preceding equality, we use mathematical induction. Since we've already shown the equation to be true for i=1,2, we take as the induction hypothesis that it is true whenever i≤j<n. Now we must prove that it is true for j+1. Using **Equation (5.4)** yields

mj+1=2mj−mj−1−2=2j(m1−j+1)−(j−1)(m1−j+2)−2 (by the induction hypothesis)=(j+1)m1−2j2+2j+j2−3j+2−2=(j+1)m1−j2−j =(j+1)(m1−j)

which completes the induction proof of **(5.5)**. Letting i=n in **(5.5)**, and using that mn=0, now yields that

m1=n−1

which, again using **(5.5)**, gives the result

mi=i(n−i)

Thus, the mean number of games played when there are only *2* players with initial amounts i and n−i is the product of their initial amounts. Because both players play all stages, this is also the mean number of stages involving player 1.

Now let us return to the problem involving r players with initial amounts ni,i=1,...∑i=1r ni=n. Let X denote the number of stages needed to obtain a victor, and let Xi denote the number of stages involving player i. Now, from the point of view of player i, starting with ni, he will continue to play stages, independently being equally likely to win or lose each one, until his fortune is either n or 0. Thus, the number of stages he plays is exactly

the same as when he has a single opponent with an initial fortune of n−ni. Consequently, by the preceding result, it follows that

$$E[X_i]=n_i(n-n_i)$$

so

$$E[\sum_{i=1}^{r}X_i]=\sum_{i=1}^{r}n_i(n-n_i)=n^2-\sum_{i=1}^{r}n_i^2$$

But because each stage involves two players,

$$X=12\sum_{i=1}^{r}X_i$$

Taking expectations now yields

$$E[X]=12(n^2-\sum_{i=1}^{r}n_i^2)$$

It is interesting to note that while our argument shows that the mean number of stages does not depend on the manner in which the teams are selected at each stage, the same is not true for the distribution of the number of stages. To see this, suppose r=3, n1=n2=1, and n3=2. If players *1* and *2* are chosen in the first stage, then it will take at least three stages to determine a winner, whereas if player 3 is in the first stage, then it is possible for there to be only two stages.

In our next example, we use conditioning to verify a result previously noted in **Section 6.3.1**: that the expected number of uniform (0,1) random variables that need to be added for their sum to exceed 1 is equal to e.

Example 5j

Let U1,U2,… be a sequence of independent uniform (0, 1) random variables. Find E[N] when

$$N=\min\{n: \sum_{i=1}^{n}U_i>1\}$$

Solution

We will find E[N] by obtaining a more general result. For $x\in[0,1]$, let

$$N(x)=\min\{n: \sum_{i=1}^{n}U_i>x\}$$

and set

$$m(x)=E[N(X)]$$

That is,N(x) is the number of uniform (0, 1) random variables we must add until their sum exceeds x, and m(x) is its expected value. We will now derive an equation for m(x) by conditioning on U1. This gives, from **Equation (5.1b)**,

$$m(x)=\int_0^1 E[N(X)|U_1=y]\,dy \tag{5.6}$$

Now,

$$E[N(X)|U_1=y]=\begin{cases}1 & \text{if } y>x\\ 1+m(x-y) & \text{if } y\le x\end{cases} \tag{5.7}$$

The preceding formula is obviously true when y>x. It is also true when y≤x, since, if the first uniform value is y, then, at that point, the remaining number of uniform random variables needed is the same as if we were just starting and were going to add uniform random variables until their sum exceeded x−y. Substituting **Equation (5.7)** into **Equation (5.6)** gives

$$m(x)=1+\int_0^x m(x-y)\,dy=1+\int_0^x m(u)\,du \qquad \text{by letting } u=x-y$$

Differentiating the preceding equation yields

$$m'(x)=m(X)$$

or, equivalently,

$$m'(x)m(x)=1$$

Integrating this equation gives

$$\log[m(X)]=x+c$$

or

$$m(x)=kex$$

Since $m(0)=1$, it follows that $k=1$, so we obtain

$$m(x)=ex$$

Therefore,$m(1)$, the expected number of uniform (0, 1) random variables that need to be added until their sum exceeds 1, is equal to e.

7.5.3 Computing probabilities by conditioning

Not only can we obtain expectations by first conditioning on an appropriate random variable, but we can also use this approach to compute probabilities. To see this, let A denote an arbitrary event, and define the indicator random variable X by

$$X=\{\ 1\text{if }\ A\ \text{occurs}0\text{if }\ A\ \text{does not occur}$$

It follows from the definition of X that

$$E[X]=P(A)E[X|Y=y]=P(A|Y=y)\quad \text{for any random variable }\ Y$$

Therefore, from **Equations (5.1a**) and **(5.1b)**, we obtain

$$P(A)=\sum yP(A|Y=y)P(Y=y)\quad \text{if }\ Y\ \text{is discrete}=\int -\infty\infty P(A|Y=y)fY(y)dy\quad \text{if }\ Y\ \text{is continuous} \tag{5.8}$$

Note that if Y is a discrete random variable taking on one of the values $y1,\ldots,yn$, then by defining the events $Bi,i=1,\ldots,n$, by $Bi=\{Y=yi\}$, **Equation (5.8)** reduces to the familiar equation

$$P(A)=\sum i=1nP(A|Bi)P(Bi)$$

where $B1,\ldots,Bn$ are mutually exclusive events whose union is the sample space.

Example 5k The best-prize problem

Suppose that we are to be presented with n distinct prizes, in sequence. After being presented with a prize, we must immediately decide whether to accept it or to reject it and consider the next prize. The only information we are given when deciding whether to accept a prize is the relative rank of that prize compared to ones already seen. That is, for instance, when the fifth prize is presented, we learn how it compares with the four prizes we've already seen. Suppose that once a prize is rejected, it is lost, and that our objective is to maximize the probability of obtaining the best prize. Assuming that all n! orderings of the prizes are equally likely, how well can we do?

Solution

Rather surprisingly, we can do quite well. To see this, fix a value $k,0\le k<n$, and consider the strategy that rejects the first k prizes and then accepts the first one that is better than all of those first k. Let $Pk(\ \text{best}\ )$ denote the probability that the best prize is selected when this strategy is employed. To compute this probability, condition on X, the position of the best prize. This gives

$$Pk(\ \text{best}\ )=\sum i=1nPk(\ \text{best}\ |X=i)P(X=i)$$

$$=1n\sum i=1nPk(\ \text{best}\ |X=i)$$

Now, on the one hand, if the overall best prize is among the first k, then no prize is ever selected under the strategy considered. That is,

$$P_k(\text{ best }|X=i)=0 \quad \text{if } i\leq k$$

On the other hand, if the best prize is in position i, where i>k, then the best prize will be selected if the best of the first i−1 prizes is among the first k(for then none of the prizes in positions k+1,k+2,…,i−1 would be selected). But, conditional on the best prize being in position i, it is easy to verify that all possible orderings of the other prizes remain equally likely, which implies that each of the first i−1 prizes is equally likely to be the best of that batch. Hence, we have

$$P_k(\text{ best }|X=i)=P\{\text{best of first } i-1 \text{ is among the first } k|X=i\}=\frac{k}{i-1} \quad \text{if } i>k$$

From the preceding, we obtain

$$P_k(\text{best })=\frac{k}{n}\sum_{i=k+1}^{n}\frac{1}{i-1}\approx\frac{k}{n}\int_{k+1}^{n}\frac{1}{x-1}\,dx=\frac{k}{n}\log\left(\frac{n-1}{k}\right)\approx\frac{k}{n}\log\left(\frac{n}{k}\right)$$

Now, if we consider the function

$$g(x)=x\log\left(\frac{n}{x}\right)$$

then

$$g'(x)=\log\left(\frac{n}{x}\right)-1$$

so

$$g'(x)=0\Rightarrow\log\left(\frac{n}{x}\right)=1\Rightarrow x=\frac{n}{e}$$

Thus, since $P_k(\text{ best })\approx g(k)$, we see that the best strategy of the type considered is to let the first *n/e* prizes go by and then accept the first one to appear that is better than all of those. In addition, since $g(n/e)=1/e$, the probability that this strategy selects the best prize is approximately $1/e\approx.36788$.

Remark Most people are quite surprised by the size of the probability of obtaining the best prize, thinking that this probability would be close to 0 when n is large. However, even without going through the calculations, a little thought reveals that the probability of obtaining the best prize can be made reasonably large. Consider the strategy of letting half of the prizes go by and then selecting the first one to appear that is better than all of those. The probability that a prize is actually selected is the probability that the overall best is among the second half, and this is $\frac{1}{2}$. In addition, given that a prize is selected, at the time of selection that prize would have been the best of more than n/2 prizes to have appeared and would thus have probability of at least $\frac{1}{2}$ of being the overall best. Hence, the strategy of letting the first half of all prizes go by and then accepting the first one that is better than all of those prizes has a probability greater than $\frac{1}{4}$ of obtaining the best prize.

Example 5l

Let U be a uniform random variable on (0, 1), and suppose that the conditional distribution of X, given that U=p, is binomial with parameters n and p. Find the probability mass function of X.

Solution

Conditioning on the value of U gives

$$P\{X=i\}=\int_0^1 P\{X=i|U=p\}f_U(p)\,dp=\int_0^1 P\{X=i|U=p\}\,dp=\frac{n!}{i!(n-i)!}\int_0^1 p^i(1-p)^{n-i}\,dp$$

Now, it can be shown (a probabilistic proof is given in **Section 6.6)** that

$$\int_0^1 p^i(1-p)^{n-i}dp=\frac{i!(n-i)!}{(n+1)!}$$

Hence, we obtain

$$P\{X=i\}=\frac{1}{n+1} \quad i=0,\ldots,n$$

That is, we obtain the surprising result that if a coin whose probability of coming up heads is uniformly distributed over (0, 1) is flipped n times, then the number of heads occurring is equally likely to be any of the values 0,…,n.

Because the preceding conditional distribution has such a nice form, it is worth trying to find another argument to enhance our intuition as to why such a result is true. To do so, let U,U1,…,Un be n+1 independent uniform (0, 1) random variables, and let X denote the number of the random variables U1,…,Un that are smaller than U.Since all the random variables U,U1,…,Un have the same distribution, it follows that U is equally likely to be the smallest, or second smallest, or largest of them; so X is equally likely to be any of the values 0,1,…,n. However, given that U=p, the number of the Ui that are less than U is a binomial random variable with parameters n and p, thus establishing our previous result.

Example 5m

A random sample of X balls is chosen from an urn that contains n red and m blue balls. If X is equally likely to be any of the values 1,…,n, find the probability that all the balls in the sample are red.

Solution

Conditioning on X yields

$$P(\text{ all balls are red })=\sum_{i=1}^{n} P(\text{ all balls are red } | X=i) P(X=i)$$

Now, given that the sample is of size i, each of the (n+mi) subsets of size i is equally likely to be the chosen set of balls. As (ni) of these subsets have all red balls, it follows that P{ all balls are red|X=i }=(ni)(n+mi) and thus that

$$P(\text{all balls are red})=\frac{1}{n}\sum_{i=1}^{n}\frac{\binom{n}{i}}{\binom{n+m}{i}}$$

However, though not obvious, it turns out that the preceding can be simplified, and indeed yields the surprising result that

$$P(\text{ all balls are red })=\frac{1}{m+1}, \quad \text{for all } n,m$$

To prove the preceding formula, we will not make use of our earlier result, but rather we will use induction on n. When n=1, the urn contains 1 red and m blue balls and so a random sample of size 1 will be red with probability 1m+1. So, assume the result is true whenever the urn contains n−1 red and m blue balls and a random sample whose size is equally likely to be any of 1,…,n−1 is to be chosen. Now consider the case of n red and m blue balls. Start by conditioning not on the value of X but only on whether or not X=1. This yields

$$P(\text{ all balls are red })=P(\text{ all red } | X=1) P(X=1)+P(\text{ all red } | X>1) P(X>1)=\frac{n}{n+m}\frac{1}{n}+P(\text{ all red } | X>1)\frac{n-1}{n}$$

Now, if X>1 then in order for all balls in the sample to be red, the first one chosen must be red, which occurs with probability nn+m, and then all of the X−1 remaining balls in the sample must be red. But given that the first ball chosen is red, the remaining X−1 balls will be randomly selected from an urn containing n−1 red and m blue balls. As X−1, given that X>1, is equally likely to be any of the values 1,…,n−1, it follows by the induction hypothesis that

$$P(\text{ all balls are red } | X>1)=\frac{n}{n+m}\frac{1}{m+1}$$

Thus,

$$P(\text{ all balls are red })=\frac{1}{n+m}+\frac{n}{n+m}\frac{1}{m+1}+\frac{n-1}{n}=\frac{1}{n+m}\left(1+\frac{n-1}{m+1}\right)=\frac{1}{m+1}$$

Example 5n

Suppose that X and Y are independent continuous random variables having densities fX and fY, respectively. Compute P{X<Y}.

Solution

Conditioning on the value of Y yields

$$P\{X<Y\}=\int_{-\infty}^{\infty} P\{X<Y|Y=y\} f_Y(y)\,dy=\int_{-\infty}^{\infty} P\{X<y|Y=y\} f_Y(y)\,dy=\int_{-\infty}^{\infty} P\{X<y\} f_Y(y)\,dy \quad \text{by independence} =\int_{-\infty}^{\infty} F_X(y) f_Y(y)\,dy$$

where

$$F_X(y)=\int_{-\infty}^{y} f_X(x)\,dx$$

Example 5o

Suppose that X and Y are independent continuous random variables. Find the distribution function and the density function of X+Y.

Solution

By conditioning on the value of Y, we obtain

$$P\{X+Y<a\}=\int_{-\infty}^{\infty} P\{X+Y<a|Y=y\}\, f_Y(y)\,dy=\int_{-\infty}^{\infty} P\{X+y<a|Y=y\}\, f_Y(y)\,dy=\int_{-\infty}^{\infty} P\{X<a-y\}\, f_Y(y)\,dy=\int_{-\infty}^{\infty} F_X(a-y)\, f_Y(y)\,dy$$

Differentiation yields the density function of X+Y:

$$f_{X+Y}(a)=\frac{d}{da}\int_{-\infty}^{\infty} F_X(a-y)\, f_Y(y)dy=\int_{-\infty}^{\infty}\frac{d}{da}F_X(a-y)\, f_Y(y)dy=\int_{-\infty}^{\infty} f_X(a-y)\, f_Y(y)dy$$

7.5.4 Conditional variance

Just as we have defined the conditional expectation of X given the value of Y, we can also define the conditional variance of X given that Y=y:

$$\text{Var}(X|Y)\equiv E[(X-E[X|Y])^2|Y]$$

That is, Var (X|Y) is equal to the (conditional) expected square of the difference between X and its (conditional) mean when the value of Y is given. In other words, Var (X|Y) is exactly analogous to the usual definition of variance, but now all expectations are conditional on the fact that Y is known.

There is a very useful relationship between Var (X), the unconditional variance of X, and Var (X|Y), the conditional variance of X given Y, that can often be applied to compute Var (X). To obtain this relationship, note first that by the same reasoning that yields Var (X)=E[X2]−(E[X] stretchy='false')2, we have

$$\text{Var}(X|Y)=E[X^2|Y]-(E[X|Y])^2$$

so

$$E[\text{Var}(X|Y)]=E[E[X^2|Y]]-E[(E[X|Y])^2]=E[X^2]-E[(E[X|Y])^2] \tag{5.9}$$

Also, since E[E[X|Y]]=E[X], we have

$$\text{Var}(E[X|Y])=E[(E[X|Y])^2]-(E[X])^2 \tag{5.10}$$

Hence, by adding **Equations (5.9)** and **(5.10)**, we arrive at the following proposition.

Proposition 5.2 The Conditional Variance Formula

$$\text{Var}(X)=E[\text{Var}(X|Y)]+\text{Var}(E[X|Y])$$

Example 5p

Suppose that by any time t the number of people who have arrived at a train depot is a Poisson random variable with mean λt. If the initial train arrives at the depot at a time (independent of when the passengers arrive) that is uniformly distributed over (0,T), what are the mean and variance of the number of passengers who enter the train?

Solution

For each t≥0, let N(t) denote the number of arrivals by t, and let Y denote the time at which the train arrives. The random variable of interest is then N(Y). Conditioning on Y gives

$E[N(Y)|Y=t]=E[N(t)|Y=t]=E[N(t)]$ by the independence of Y and $N(t)=\lambda t$ since $N(t)$ is Poisson with mean λt

Hence,

$E[N(Y)|Y]=\lambda Y$

so taking expectations gives

$E[N(Y)]=\lambda E[Y]=\lambda T2$

To obtain $Var(N(Y))$, we use the conditional variance formula:

$Var(N(Y)|Y=t)=Var(N(t)|Y=t)=Var[N(t)]$ by independence$=\lambda t$

Thus,

$Var(N(Y)|Y)=\lambda Y E[N(Y)|Y]=\lambda Y$

Hence, from the conditional variance formula,

$Var(N(Y))=E[\lambda Y]+Var(\lambda Y)=\lambda T2+\lambda 2T212$

where we have used the fact that $Var(Y)=T2/12$.

Example 5q Variance of a sum of a random number of random variables

Let $X1,X2,\ldots$ be a sequence of independent and identically distributed random variables, and let N be a nonnegative integer-valued random variable that is independent of the sequence $Xi, i\geq 1$. To compute $Var(\sum i=1NXi)$, we condition on N:

$E[\sum i=1 N X i |N] = NE[X] \; Var(\sum i=1 N X i |N) = N \, Var(X)$

The preceding result follows because, given N, $\sum i=1NXi$ is just the sum of a fixed number of independent random variables, so its expectation and variance are just the sums of the individual means and variances, respectively. Hence, from the conditional variance formula,

$Var(\sum i=1NXi)=E[N] \; Var(X)+(E[X])2 \; Var(N)$